Random Forest Machine Learning Project

Dataset Name

Concrete Compressive Strength Dataset (Kaggle). The dataset contains 1,030 records and 9 columns, with 8 input features and 1 target variable.

Problem Statement

To build a Random Forest Regression model that predicts the compressive strength of concrete using its material composition and age.

Data Preprocessing Steps

The dataset was loaded and explored using Pandas. Missing values and duplicate records were checked. Column-name spaces were cleaned, and the dataset was divided into input features (X) and the target variable (y). The data was then split into 80% training data and 20% testing data.

Model Used

Random Forest Regressor with 100 decision trees (n_estimators = 100) and random_state = 42.

Evaluation Metrics

Metric	Result
MAE	3.4979 MPa
MSE	27.6173
RMSE	5.2552 MPa
R^2 Score	0.9074

Key Findings

The model achieved an R^2 score of 0.9074, indicating that it explains approximately 90.74% of the variation in concrete compressive strength on the test dataset. The MAE of 3.50 MPa and RMSE of 5.26 MPa indicate relatively small prediction errors.

Conclusion

The Random Forest Regression model successfully predicted concrete compressive strength with good performance on the test data. The results show that Random Forest is an effective model for this regression problem.